Iterative Repair with Weak Verifiers for Few-shot Transfer in KBQA with Unanswerability

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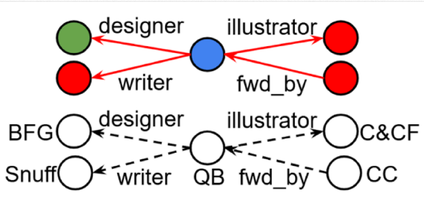
Knowledge Base Question Answering



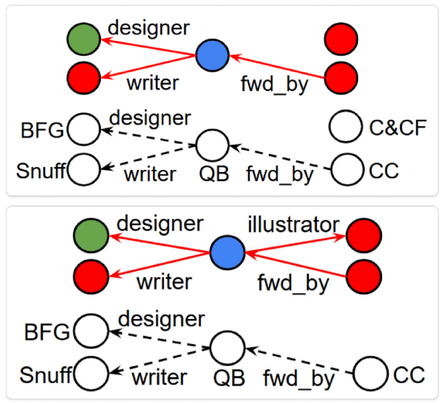
Question: Which book was illustrated by Quentin Blake?

Logical form: QB illustrator ?x

Answer: {C&CF}

- KBs commonly used by large enterprises for **storing and access semantic facts**
- Two layers: **Schema** & **Data**

- KBQA: Answer natural language question using KB
- **KBQA as semantic parsing**
  - Generate logical form (LF)
  - Execute over KB to retrieve answer



KB may not have answer

- Schema incompleteness
- Data incompleteness

KBQA with Unanswerability

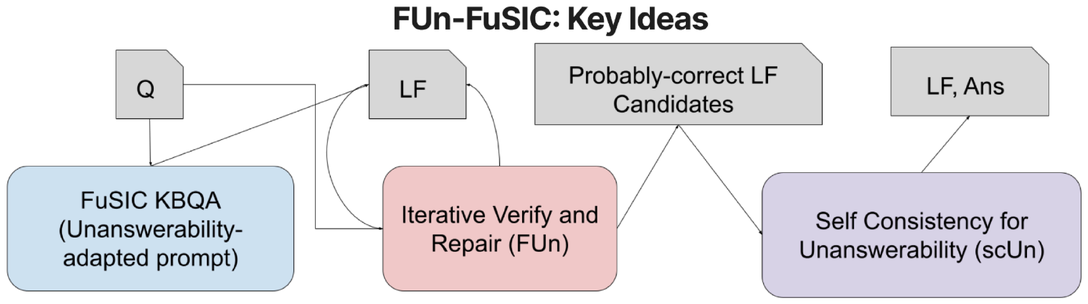
- Detect if KB admits valid LF
  - Schema-level check
- Execute LF to retrieve answer
  - Data-level check

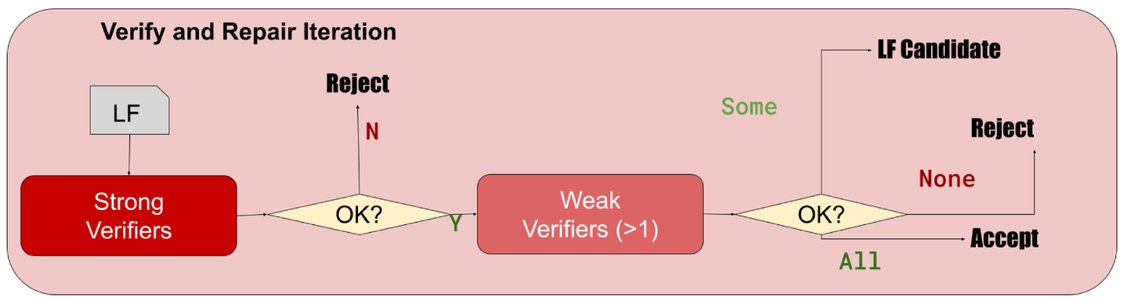
KBQA Deployment Challenges: Robustness vs Volume of Labeled Training Data

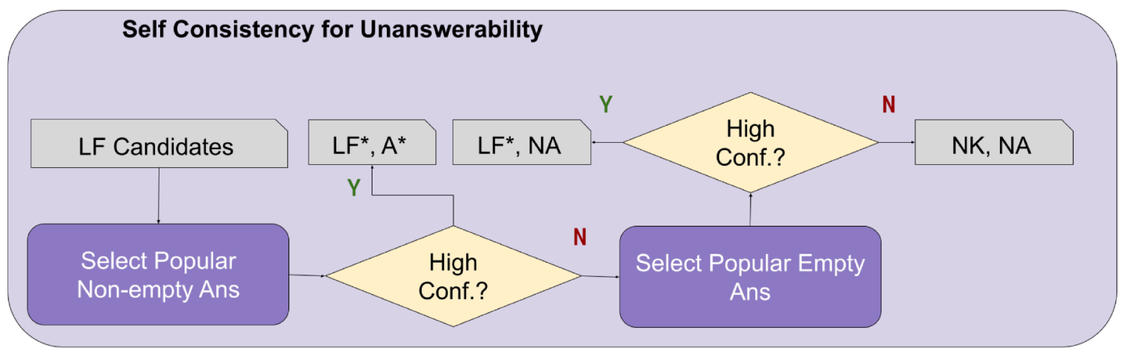
- **KBQA with Unanswerability Detection**: Multi-stage DL architectures + thousands of labeled training examples
- **KBQA Few-shot** (In-domain & Transfer Learning: Cannot detect unanswerability

- **Novel KBQA Task:** Few-shot transfer learning with unanswerability detection
- **In-domain:** Tens of few-shot examples (Answerable + Unanswerable)
- **Transfer setting:** Source domain with thousands of labeled training examples (only Answerable)

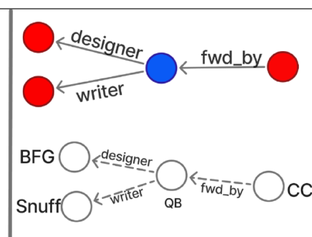
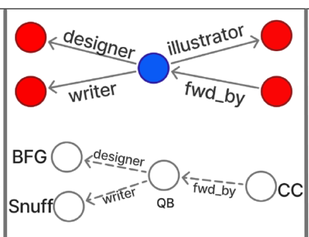
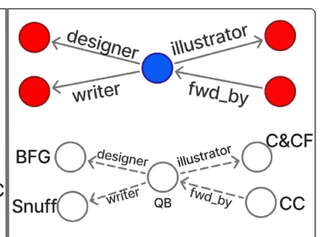
FUn-FuSIC: Key Ideas







FUn-FuSIC: Illustration

Q. Which book was illustrated by Quentin Blake?	 KB1: Schema level Unanswerable	 KB2: Data level Unanswerable	 KB3: Answerable
FUn i=1 **V2:** KB INCONSISTENCY	**LF1:** ?x{?x writer QB} True	**LF1:** ?x{?x writer QB} True	**LF1:** ?x{?x writer QB} True
FUn i=2 **V2:** KB INCONSISTENCY **V3:** Q-LF DISAGREEMENT **V4b:** ANS INCONSISTENCY	**LF2:** {QB writer ?x} False True False(Snuff)	**LF2:** {QB writer ?x} False True False(Snuff)	**LF2:** {QB writer ?x} False True False(Snuff)
FUn i=3 **V2:** KB INCONSISTENCY **V3:** Q-LF DISAGREEMENT **V4b:** ANS INCONSISTENCY	**LF3:** {?x fwd_by QB} False True False(CC)	LF3: {QB illustrator ?x} False False True ( )	**LF3:** {QB illustrator ?x} False False False(C&CF) Return (LF3,{C&CF})
FUn i=4 **V2:** KB INCONSISTENCY **V3:** Q-LF DISAGREEMENT **V4b:** ANS INCONSISTENCY	**LF4:** {QB designer ?x} False True False(BFG)	**LF4:** {QB designer ?x} False True False (BFG)	
scUN Non-empty Ans - Best LF Empty Ans - Best LF	L = {LF2, LF3, LF4} {LF2, LF3, LF4} - No consensus { }: No Consensus; Return (NK,NA)	L = {LF2, LF3, LF4} {LF2, LF4} - No consensus {LF3} : Consensus; Return (LF3, NA)	

Top Graph: schema: classes (book & person) & relations **Bottom Graph:** data
Gray blocks: 4 iterations of FUn (i = 1 to 4) **Blue block:** scUn; L is the set of logical forms
Verifiers: Strong and Weak; return True or False **Output:** Logical Form(or NK), Answer(or NA)

Datasets

- 2 new datasets for Few-shot KBQA Transfer w. Unanswerability
  - WebQSP → GrailQAbility (AUD1), WebQSP → GraphQAbility (AUD2)
- 2 existing datasets for Few-shot KBQA Transfer w. only Answerable
  - WebQSP → GrailQA-Tech (AD1), WebQSP → GraphQA-Pop (AD2)

Evaluation Metrics

- Answer: F1
- Logical Form: EM-s

Results

Few-shot Transfer with Unanswerability

Model	AUD1										AUD2									
	Overall		Answerable		Unanswerable		Schema-U		Data-U		Overall		Answerable		Unanswerable		Schema-U		Data-U	
	F1	EM-s	F1	EM-s	F1	EM-s	F1	EM-s	F1	EM-s	F1	EM-s	F1	EM-s	F1	EM-s	F1	EM-s	F1	EM-s
RetinaQA	58.4	42.2	28.7	26.0	**88.0**	58.4	94.1	79.4	76.3	14.1	49.7	35.8	18.7	15.2	80.7	56.4	83.2	72.3	73.7	12.1
Pangu	54.5	43.8	31.2	29.6	83.8	58.0	91.1	**87.9**	69.6	0.0	53.4	33.0	30.3	26.4	76.5	39.6	77.3	74.4	73.3	0.0
KB-Binder	43.7	33.0	19.5	16.5	67.9	49.5	75.1	70.1	53.1	9.5	44.3	36.1	27.5	21.6	61.0	50.7	67.0	60.9	41.2	6.8
Fusic-U	76.6	48.2	**67.5**	59.2	85.6	37.2	85.4	30.9	**86.0**	**49.4**	67.5	34.8	49.3	40.0	85.7	29.6	86.6	19.0	**83.3**	**51.5**
FUn-Fusic	**76.6**	**60.2**	67.1	**61.2**	85.1	**59.2**	**85.8**	70.9	83.8	36.5	**70.0**	**53.8**	**50.7**	**42.8**	**89.2**	**64.8**	**92.4**	**75.6**	80.3	34.8

1. FUn-FuSIC has overall SoTA performance for both F1 and EM-s across both newly constructed datasets with unanswerability
2. FUn-FuSIC has robust performance across both answerable & unanswerable questions, as well as different categories of unanswerability

With only Answerable Datasets

Model	AD1	AD2
W knwl of answerability (F1)		
FuSIC	70.8	52.3
FUn-FuSIC (sc)	**73.6**	**67**
W/o knwl of answerability(F1)		
FuSIC	62.6	43.4
FUn-FuSIC (scUn)	**71.2**	**65**

1. FUn-FuSIC beats prior SoTA on existing transfer datasets

References

1. Few-shot transfer learning for knowledge base question answering: Fusing supervised models with in-context learning, Patidar et al, ACL 2024
2. "RETINAQA : A knowledge base question answering model robust to both answerable and unanswerable questions", Faldu et al, ACL 2023
3. "Do I have the Knowledge to Answer? Investigating Answerability of Knowledge Base Questions", Patidar et al, ACL 2023